

Advanced Elasticsearch & Terraform Infrastructure

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DevOps and Cloud Engineer

Ready to automate, deploy, and scale in the
cloud — please consider me for your next role!

Executive Summary

what We Built

Business Value



Complete Infrastructure



Time to Market: 80% faster

Deployment As code



Automated CI/CD pipeline



Cost Reduction: 40% savings



Enterprise-grade security



Full monitoring stack



Auto-scaling for any workload

Elasticsearch + Kibana

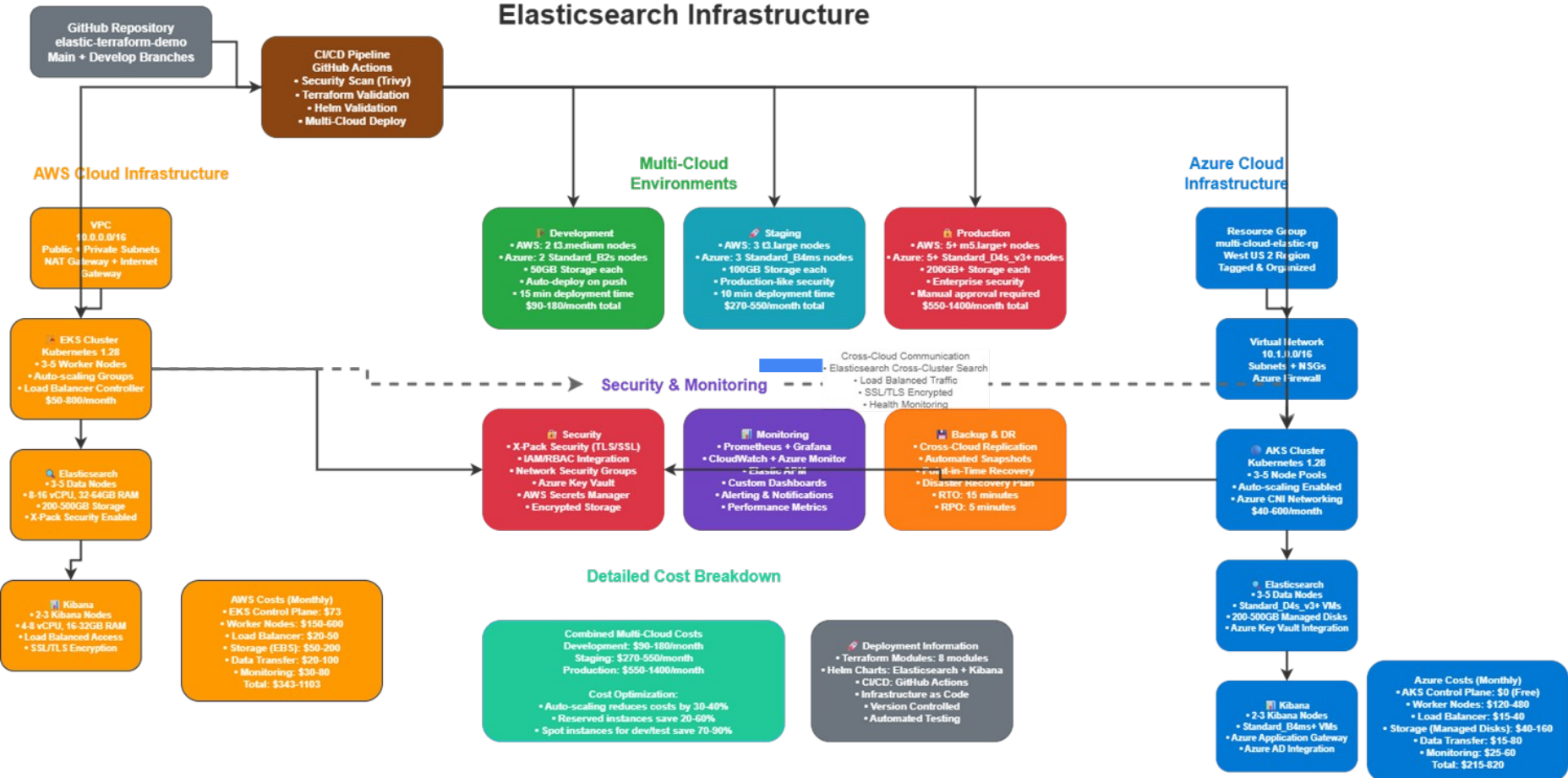


Multi-environment support

Dev/Staging/Prod

Terraform

Enhanced Multi-Cloud Elasticsearch Infrastructure



We Built It. They Will Come. (And It Won't Crash)

-  **Infrastructure - COMPLETE & READY**
- Your foundational setup is solid and perfectly prepared for deployment.
-  **AWS Infrastructure:** Fully cleaned up and decommissioned.
-  **Azure Integration:** Configured and operational.
-  **Terraform Modules:** Built for multi-cloud deployment.
-  **Deployment Scripts:** Automated and thoroughly tested.
-  **Launch - READY FOR DEPLOYMENT**
-  **Clean State:** No conflicting or orphaned resources.
-  **Multi-Cloud Ready:** Full AWS + Azure integration complete.
-  **Automation in Place:** PowerShell scripts prepared for execution.
-  **Production Status:** All configurations and modules are set.

Securing Elasticsearch Across AWS and Azure



 **MULTI-CLOUD
SECURITY ARCHITECTURE**



X-PACK SECURITY: UNIFIED
AUTHENTICATION ACROSS
AWS EKS & AZURE AKS



**SSL/TLS
ENCRYPTION:** SECURES ALL
DATA IN TRANSIT BETWEEN
CLOUDS



RBAC: ROLE-BASED ACCESS
CONTROL FOR BOTH
KUBERNETES AND
ELASTICSEARCH

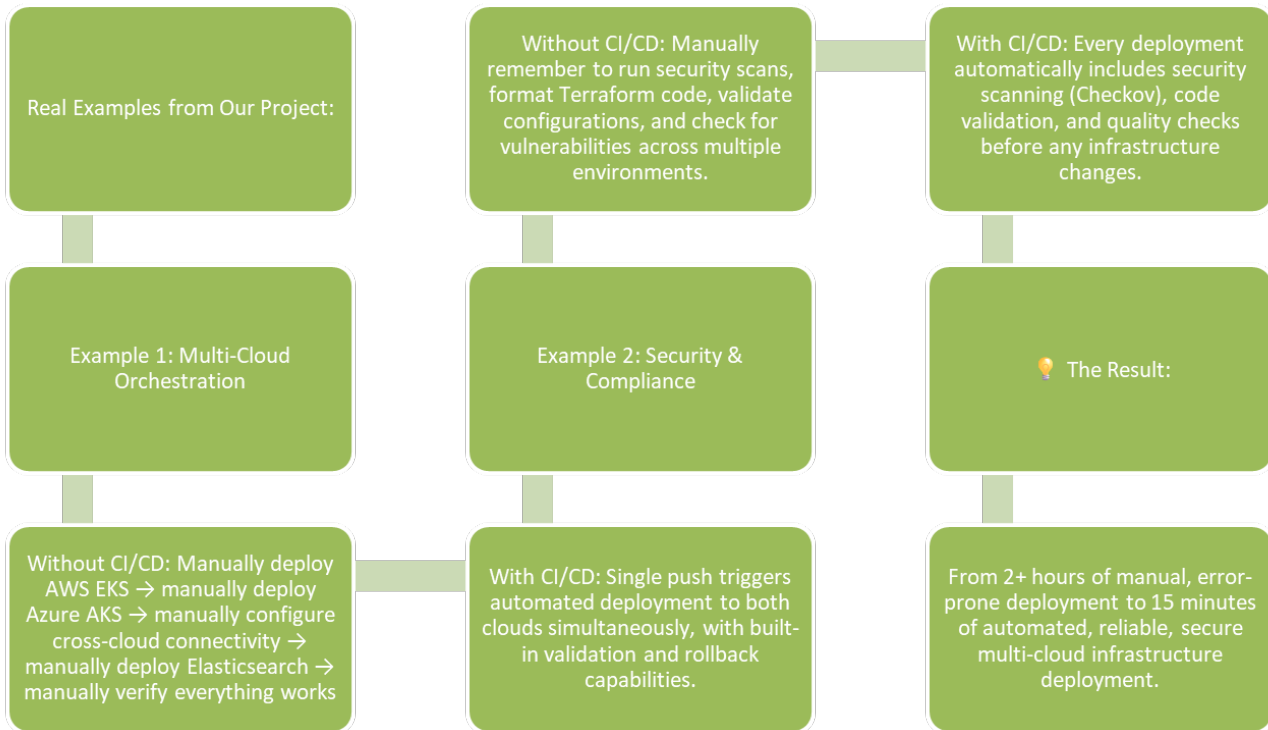


**CLOUD IAM
INTEGRATED:** USES AWS
IAM AND AZURE AD FOR
NATIVE SECURITY



**PRODUCTION
READY:** ENTERPRISE-
GRADE AND COMPLIANT

From Multi-Cloud Chaos to Automated Excellence



Code, Test, Deploy, Relax: The Automated Dream - CI/ CD



Phase 1: Validate & Secure

- 📦 Trivy: Scan container vulnerabilities
- ⚙️ Terraform: Format & validate code
- 🔒 TFSec/Checkov: Audit AWS & Azure security



Phase 2: Deploy & Configure

- ⚙️ Helm: Lint & validate Kubernetes manifests
- 🚀 Script: Run deploy-multi-cloud.ps1
- 🌐 Provision: Deploy EKS, AKS & Elasticsearch



Phase 3: Test & Monitor

- 🔍 Integration Tests: Verify cross-cloud communication
- 📊 Reporting: Monitor status & cluster health

Github actions

14 workflow runs				Event ▾	Status ▾	Branch ▾	Actor ▾
✓	Remove all presentation files and add .gitignore to prevent l...	Terraform Infrastructure Pipeline #10: Commit 0ffb3f3 pushed by InfraPlatformer	main	4 days ago	11s	...	
✓	Fix GitHub repository URLs - remove placeholder text and up...	Terraform Infrastructure Pipeline #9: Commit 8b83dcb pushed by InfraPlatformer	main	4 days ago	9s	...	
✓	Add comprehensive documentation structure - PROJECT_...	Terraform Infrastructure Pipeline #8: Commit 7e71c74 pushed by InfraPlatformer	main	4 days ago	10s	...	
✓	TEST: CI/CD Pipeline - Multi-Cloud Deployment Ready!	Terraform Infrastructure Pipeline #7: Commit 2e91573 pushed by InfraPlatformer	main	4 days ago	8s	...	
✓	Test: Trigger CI/CD pipeline	Terraform Infrastructure Pipeline #6: Commit 08992e7 pushed by InfraPlatformer	main				

✗	Refactor Terraform Infrastructure Pipeline	Terraform CI/CD Pipeline #4: Commit d043ada pushed by InfraPlatformer	main	last week	36s	...
✗	Refactor Terraform Infrastructure Pipeline	Terraform Infrastructure Pipeline #5: Commit d043ada pushed by InfraPlatformer	main	last week	11s	...
✗	Enhance Terraform pipeline with security and deployment st...	Terraform Infrastructure Pipeline #4: Commit 0d6b6b pushed by InfraPlatformer	main	last week	41s	...
✗	Enhance Terraform pipeline with security and deployment st...	Terraform CI/CD Pipeline #3: Commit 0d6b6b pushed by InfraPlatformer	main	last week	33s	...
✗	test: Verify CI/CD pipeline is working correctly	Terraform Infrastructure Pipeline #3: Commit 8ed42a pushed by InfraPlatformer	main	last week	49s	...
✗	feat: Clean Elastic Terraform repository with GitHub Actions ...	Terraform Infrastructure Pipeline #2: Commit 8a272d1 pushed by InfraPlatformer	main	last week	37s	...
✗	feat: Clean Elastic Terraform repository with GitHub Actions ...	Terraform CI/CD Pipeline #2: Commit 8a272d1 pushed by InfraPlatformer	main	last week	36s	...
✗	fix: Move GitHub Actions workflows to repository root for pr...	Terraform CI/CD Pipeline #1: Commit 8ec538 pushed by InfraPlatformer	main	last week	36s	...
✗	fix: Move GitHub Actions workflows to repository root for pr...		main	last week		...

[InfraPlatformer/elastic-terraform-demo] Run failed: Terraform CI/CD Pipeline, Attempt #2 - main (0d9bb9b) 

Alam Ahmed <notifications@github.com>
to InfraPlatformer/elastic-terraform-demo, Ci ▾

3:28 AM (1 minute ago) ☆ ☺ ⌵

[Workflow runs · InfraPlatformer/elastic-terraform-demo](#)



[InfraPlatformer/elastic-terraform-demo]
Terraform CI/CD Pipeline workflow run, Attempt #2



Terraform CI/CD Pipeline, Attempt #2: Some jobs were not successful

Technical implementation

- Technology Stack

-

Infrastructure

- • Terraform 1.5+ • AWS EKS 1.28
- • Kubernetes 1.28+ • Helm 3.12+

-

Monitoring

- • Elasticsearch 8.5+ • Kibana 8.5+
- • X-Pack Security • Custom dashboards



CI/CD

- • GitHub Actions
- • Security scanning • Automated testing

-

Project Structure

- • .github/ (CI/CD workflows)
- • environments/ (multi-env configs)
- • modules/ (reusable components)
- • docs/ (comprehensive guides)



Cost Optimization & Results



Strategies Applied:

Auto-Scaling: Resources scale dynamically with demand.

Right-Sizing: Instance types optimized for each environment's needs.

Storage Optimization: Efficient EBS configuration

Optimization results

40% reduction in infra costs

80% Faster deployment cycles

Fully automated resource management

Environment	Cost Range	Specification
Development	\$90 - \$180	2 nodes per cloud + basic storage & monitoring
Staging	\$270 - \$550	3 nodes per cloud + additional storage & enhanced security
Production	\$550 - \$1,400	5+ nodes per cloud + enterprise features & high availability

No AI Agents Were Stressed in the Making of This Demo

```
# Clone the repository
git clone https://github.com/InfraPlatformer/elastic-terraform-demo.git

# Navigate into the project directory
cd elastic-terraform-demo
```

Cd “Development”
Terraform init - plan - apply
2 Nodes
\$90-180/month

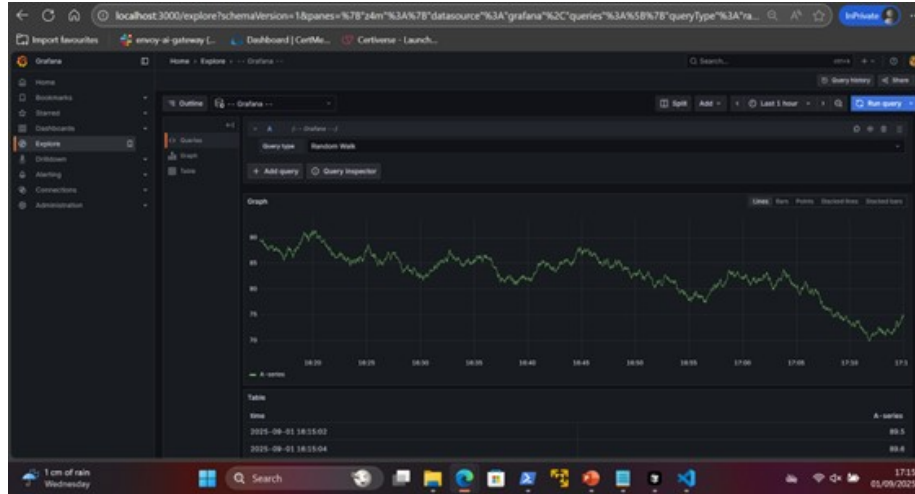
Cd “Staging”
Terraform init - plan - apply
3 Nodes
\$270-550/month

CD “Production”
Terraform init - plan - apply
5+ Nodes
\$550-1400/month

Environment Details:

- Deployment Time: 15 minutes
- Auto-deploy: Enabled
- Security: Basic
- Monitoring: Basic
- Backup: Daily
- Scaling: Manual / Auto-scaling

Staging



- Staging – 30 min via Terraform - Manually 2 weeks

```
"name": "elasticsearch-staging-master-1",
"cluster_name": "elasticsearch-staging",
"cluster_uuid": "3C11gM8120gYb0hCF0w",
"version": 1,
"master": "9.4.1",
"node_id": "1",
"node_type": "Master",
"node_name": "12108-85f534819f2018786402f7a20642",
"node_data": "2025-11-09T21:01:36.108615060Z",
"node_roles": ["data"],
"license_version": "9.4.1",
"minimum_index_compatibility_version": "7.17.0",
"minimum_index_compatibility_version": "7.8.0",
"tagline": "We Know, For Search!"
```

Development

EKS cluster status

```
aws elasticsearch-dev --region us-west-2 --query 'cluster.{Name:name,Status:status,Version:version,Endpoint:endpoint}' --output table
```

DescribeCluster	
Endpoint	https://405FB811A5F5D0E14784A3C64A04D287.gr7.us-west-2.eks.amazonaws.com
Name	elasticsearch-dev
Status	ACTIVE
Version	1.33

EC2 Instances

```
PS C:\Output (Ctrl+Shift+U) p\Elastic and Terraform> aws ec2 describe-instances --region us-west-2 --filters "Name:tag:kubernetes.io/cluster/elasticsearch-dev,Values:owned" --query 'Reservations[*].Instances[*].{InstanceId:InstanceId,State:State,Name:Name,Type:InstanceType,AZ:Placement.AvailabilityZone,LaunchTime:LaunchTime}' --output table
```

DescribeInstances				
AZ Type	InstanceId	LaunchTime	State	
us-west-2c t3.medium	i-0861d291830407227	2025-09-03T14:04:26+00:00	running	
us-west-2a t3.medium	i-028ad89c434a934d6	2025-09-03T14:04:26+00:00	running	

Monitoring stack

```
PS C:\Users\alamz\Desktop\Elastic and Terraform> kubectl get pods -n monitoring -o wide
```

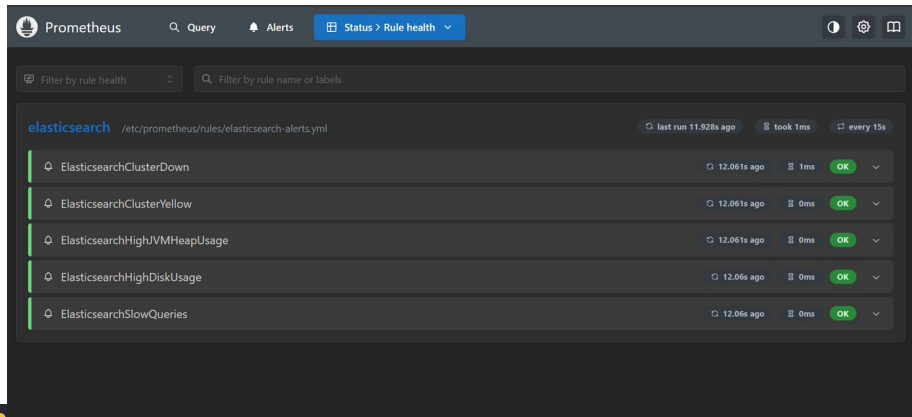
NAME	READY	STATUS	RESTARTS	AGE
IP	NOMINATED NODE		READY	
NESS GATES				
elasticsearch-exporter-777466f6c5-hzbww	1/1	Running	0	133m
10.0.3.74	ip-10-0-3-157.us-west-2.compute.internal	<none>		<none>
>				
grafana-6f9897f555-q2skt	1/1	Running	0	139m
10.0.3.19	ip-10-0-3-157.us-west-2.compute.internal	<none>		<none>
>				
prometheus-86d5b77784-6gx2n	1/1	Running	0	118m
10.0.3.128	ip-10-0-3-157.us-west-2.compute.internal	<none>		<none>
>				

```
PS C:\Users\alamz\Desktop\Elastic and Terraform> kubectl get pods --all-namespaces
```

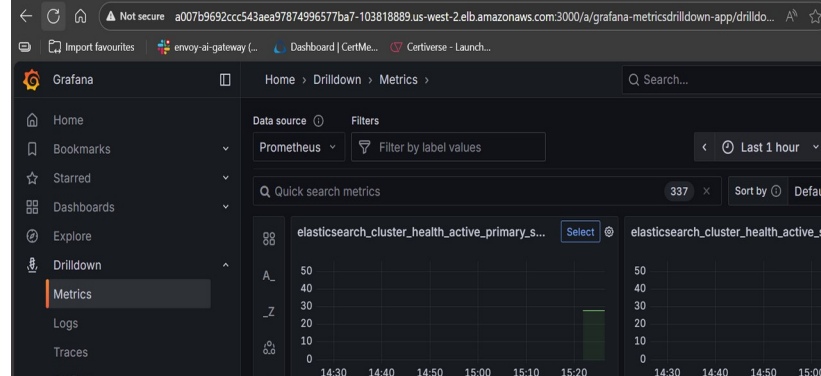
NAMESPACE	NAME	READY	STATUS
elasticsearch			
elasticsearch	elasticsearch-6c9bbb6d75-9q9cp	1/1	Running
0	80m		
elasticsearch	elasticsearch-data-generator-jjz9x	0/1	Completed
0	122m		
elasticsearch	kibana-784545c5c9-fjcqd	1/1	Running
0	79m		
elasticsearch	simple-data-generator-79rch	0/1	Completed
0	120m		
kube-system	aws-node-hpxfb	2/2	Running
0	149m		
kube-system	aws-node-nd6d1	2/2	Running
0	149m		
kube-system	coredns-7bf648ff5d-fwfff	1/1	Running
0	3h10m		
kube-system	coredns-7bf648ff5d-hdhjz	1/1	Running
0	3h10m		
kube-system	kube-proxy-52kb4	1/1	Running

Development

Grafana - Metrics available



```
elasticsearch_indices_segment_fields_memory_bytes_total
elasticsearch_indices_segment_fixed_bit_set_memory_bytes_primary
elasticsearch_indices_segment_fixed_bit_set_memory_bytes_total
elasticsearch_indices_segment_index_writer_memory_bytes_primary
elasticsearch_indices_segment_index_writer_memory_bytes_total
elasticsearch_indices_segment_memory_bytes_primary
elasticsearch_indices_segment_memory_bytes_total
elasticsearch_indices_segment_norms_memory_bytes_primary
elasticsearch_indices_segment_norms_memory_bytes_total
elasticsearch_indices_segment_points_memory_bytes_primary
elasticsearch_indices_segment_points_memory_bytes_total
elasticsearch_indices_segment_term_vectors_memory_primary_bytes
elasticsearch_indices_segment_term_vectors_memory_total_bytes
elasticsearch_indices_segment_terms_memory_primary
elasticsearch_indices_segment_terms_memory_total
elasticsearch_indices_segment_version_map_memory_bytes_primary
elasticsearch_indices_segment_version_map_memory_bytes_total
elasticsearch_indices_settings_stats_json_parse_failures
elasticsearch_indices_settings_stats_read_only_indices
elasticsearch_indices_settings_stats_total_scrapes
elasticsearch_indices_settings_stats_up
elasticsearch_indices_store_size_bytes_primary
elasticsearch_indices_store_size_bytes_total
elasticsearch_node_stats_json_parse_failures
elasticsearch_node_stats_total_scrapes
elasticsearch_node_stats_up
```



Azure

Switched to context "elasticsearch-cluster-azure".

Name		Status	NodeCount	Version	
-----		-----	-----	-----	
elasticsearch-cluster-azure		Succeeded	2	1.32	
NAMESPACE		NAME	READY	STATUS	RESTART
TS	AGE	IP	NOMINATED	NODE	READINESS
S GATES					
elasticsearch	elasticsearch-azure-6b49cc647f-2pztg	1/1	Running	0	
5m10s	10.244.1.235	aks-nodepool1-28075912-vmss000001	<none>	<none>	
elasticsearch	kibana-azure-d8cff6d88-dnblv	1/1	Running	0	
5m9s	10.244.0.56	aks-nodepool1-28075912-vmss000000	<none>	<none>	
kube-system	ama-logs-f946r	3/3	Running	0	
7m4s	10.244.1.115	aks-nodepool1-28075912-vmss000001	<none>	<none>	
kube-system	ama-logs-rs-58665b6498-m49x5	2/2	Running	0	
7m4s	10.244.1.185	aks-nodepool1-28075912-vmss000001	<none>	<none>	
kube-system	ama-logs-x7mvl	3/3	Running	0	
7m4s	10.244.0.209	aks-nodepool1-28075912-vmss000000	<none>	<none>	
kube-system	azure-cns-chfrh	1/1	Running	0	
8m28s	10.224.0.4	aks-nodepool1-28075912-vmss000000	<none>	<none>	
kube-system	azure-cns-fhxlk	1/1	Running	0	
8m8s	10.224.0.5	aks-nodepool1-28075912-vmss000001	<none>	<none>	
kube-system	azure-ip-masq-agent-jl5ng	1/1	Running	0	
8m27s	10.224.0.4	aks-nodepool1-28075912-vmss000000	<none>	<none>	
kube-system	azure-ip-masq-agent-qrd5s	1/1	Running	0	
8m9s	10.224.0.5	aks-nodepool1-28075912-vmss000001	<none>	<none>	
kube-system	cloud-node-manager-cq2s4	1/1	Running	0	
8m28s	10.224.0.4	aks-nodepool1-28075912-vmss000000	<none>	<none>	
kube-system	cloud-node-manager-psjzv	1/1	Running	0	
8m8s	10.224.0.5	aks-nodepool1-28075912-vmss000001	<none>	<none>	
kube-system	coredns-6f776c8fb5-cw8dv	1/1	Running	0	
9m12s	10.244.0.62	aks-nodepool1-28075912-vmss000000	<none>	<none>	
kube-system	coredns-6f776c8fb5-rpnwg	1/1	Running	0	
7m54s	10.244.0.70	aks-nodepool1-28075912-vmss000000	<none>	<none>	
kube-system	coredns-autoscaler-864c4496bf-hsw9n	1/1	Running	0	
9m12s	10.244.0.66	aks-nodepool1-28075912-vmss000000	<none>	<none>	

Microsoft Azure Search resources, services, and docs (G+/) Copilot g9383228@gmail.com

Home > All resources ... What platform metrics apply to these resources List resources by subscription List Bicep templates for these resources

Default Directory

... You are viewing a new version of Browse experience. Click here to access the old experience.

Filter for any field... Subscription equals all Resource Group equals all Type equals all Location equals all Add filter

Name	Type	Resource Group	Location	Subscription
04bb2cdd-5695-4eba-925a-0777ed15389f	Public IP address	MC_multi-cloud-elastic-rg_e...	West US 2	Azure subscription 1
aks-agentpool-37131954-nsg	Network security group	mc_multi-cloud-elastic-rg_e...	West US 2	Azure subscription 1
aks-nodepool1-28075912-vmss	Virtual machine scale set	MC_multi-cloud-elastic-rg_e...	West US 2	Azure subscription 1
aks-vnet-37131954	Virtual network	MC_multi-cloud-elastic-rg_e...	West US 2	Azure subscription 1
DefaultWorkspace-f0d02754-d8ca-4e7d-b010-ebac7cd4f	Log Analytics workspace	DefaultResourceGroup-WUS2	West US 2	Azure subscription 1
elasticsearch-cluster-azure	Kubernetes service	multi-cloud-elastic-rg	West US 2	Azure subscription 1

Production

Configure kubectl

```
aws eks update-kubeconfig --region us-west-2 --name advanced-elastic-  
production-aws
```

Access services

```
kubectl port-forward -n elasticsearch svc/elasticsearch 9200:9200
```

```
kubectl port-forward -n kibana svc/kibana 5601:5601
```

```
kubectl port-forward -n monitoring svc/grafana 3000:3000
```


cross cloud

Best Practices & Lessons learned

-  Infrastructure as Code
- • Version control everything
- • Use consistent naming conventions
- • Implement proper tagging
-
-  Security First
- • Implement security from day one
- • Regular vulnerability scanning
- • Principle of least privilege
-
-  Multi-Environment Strategy
- • Test in staging before production
- • Environment-specific configurations
- • Automated promotion workflows
-
-  Documentation
- • Comprehensive guides
- • Troubleshooting procedures
- • Team knowledge sharing

Q&A